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Inferences

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Q1

INFORMATION AND IDEAS

In Switzerland, the white fuzzy mountain flowers known as edelweiss are widely treated as a symbol of strength and courage. Although edelweiss can thrive in extreme conditions, they aren't notably tougher or harder to reach than other mountain flowers growing in the Swiss Alps. Historian Tobias Scheidegger has shown that the popular view of the flowers originated in the mid-1800s when mountain climbing became popular in Switzerland. Mountain climbers spread the idea that the flowers grew only in steep, icy terrains that were dangerous to climb to. Scheidegger says that these claims were self-interested. He suggests that mountain climbers presented edelweiss in this way in order to _____ blank

Which choice most logically completes the text?

- A. make themselves appear brave and strong for being able to climb to difficult places where edelweiss supposedly grew.
- B. encourage more flower enthusiasts to explore the Swiss Alps.
- C. share their observations about the unusual characteristics of edelweiss with scientists.
- D. prove that edelweiss were more common in the Swiss Alps than in other mountain regions in Europe.

The practice of logging (cutting down trees for commercial and other uses) is often thought to be at odds with forest conservation (the work of preserving forests). However, a massive study in forest management and preservation spanning 700,000 hectares in Oregon's Malheur National Forest calls that view into question. So far, results of the study suggest that forest plots that have undergone limited logging (the careful removal of a controlled number of trees) may be more robust than plots that haven't been logged at all. These results, in turn, suggest that _____ blank

Which choice most logically completes the text?

- A. logging may be useful for maintaining healthy forests, provided it is limited.
- B. other forest management strategies are more effective than limited logging.
- C. as time passes, it will be difficult to know whether limited logging has any benefits.
- D. the best way to support forest health may be to leave large forests entirely untouched.

Biologist Natacha Bodenhausen and colleagues analyzed the naturally occurring bacterial communities associated with leaves and roots of wild *Arabidopsis thaliana*, a small flowering plant. The researchers found many of the same bacterial genera in both the plants' leaves and roots. To explain this, the researchers pointed to the general proximity of *A. thaliana* leaves to the ground and noted that rain splashing off soil could bring soil-based bacteria into contact with the leaves. Alternatively, the researchers noted that wind, which may be a source of bacteria in the aboveground portion of plants, could also bring bacteria to the soil and roots. Either explanation suggests that _____ blank

Which choice most logically completes the text?

- A. bacteria carried by wind are typically less beneficial to *A. thaliana* than soil-based bacteria are.
- B. some bacteria in *A. thaliana* leaves and roots may share a common source.
- C. many bacteria in *A. thaliana* leaves may have been deposited by means other than rain.
- D. *A. thaliana* leaves and roots are especially vulnerable to harmful bacteria.

Narwhals are shy whales that live in the remote Arctic Ocean. Some of them have a long tusk, like a unicorn horn, with sensitive nerves. Narwhals are known for this tusk, but many actually don't have one and its purpose is unknown. One group of scientists came up with a possible purpose in 2014. The scientists suggested that the tusk may help narwhals determine when water around them is likely to start freezing and become dangerous for them. Marine biologist Kristin Laidre disagrees with that idea, though. She reasons that if the narwhal's tusk serves such an important purpose, then it's most likely that _____ blank

Which choice most logically completes the text?

- A. some narwhals would seek a new habitat.
- B. fewer marine animals would also have tusks.
- C. more narwhals would have a tusk.
- D. narwhals would become less shy over time.

In her 2021 article “Throwaway History: Towards a Historiography of Ephemera,” scholar Anne Garner discusses John Johnson (1882–1956), a devoted collector of items intended to be discarded, including bus tickets and campaign pamphlets. Johnson recognized that scholarly institutions considered his expansive collection of ephemera to be worthless—indeed, it wasn’t until 1968, after Johnson’s death, that Oxford University’s Bodleian Library acquired the collection, having grasped the items’ potential value to historians and other researchers. Hence, the example of Johnson serves to _____ blank

Which choice most logically completes the text?

- A. demonstrate the difficulties faced by contemporary historians in conducting research at the Bodleian Library without access to ephemera.
- B. represent the challenge of incorporating examples of ephemera into the collections of libraries and other scholarly institutions.
- C. lend support to arguments by historians and other researchers who continue to assert that ephemera holds no value for scholars.
- D. illustrate both the relatively low scholarly regard in which ephemera was once held and the later recognition of ephemera’s possible utility.

Red velvet cake has been a favorite dessert of many for years, but the recipe's origins are unclear. A bakery in Dallas, Texas, argued that it created the first recipe for the cake when the bakery opened in the 1860s. The Adams Extract Co., which sells baking products, claims to have created the recipe in the 1930s to help market their red dye. A US hotel and a Canadian department store also publicly stated that the red velvet cake sold in each of their establishments in the 1930s was an original creation, each alleging that it was the recipe author. No clear evidence has emerged to favor one of these claims over the others, however. It thus seems that _____ blank

Which choice most logically completes the text?

- A. red velvet cake was first baked sometime before the 1860s.
- B. we cannot say at present who actually baked the first red velvet cake.
- C. none of the supposed inventors of red velvet cake are likely to have invented it.
- D. the bakery in Dallas, Texas, probably invented red velvet cake.

Songbirds learn to respond to and imitate their species' songs from an early age. With each generation, small differences are introduced that result in distinct variations—called dialects—among geographically isolated populations of the same species. A research study examined whether twelve-day-old *Ficedula hypoleuca* (pied flycatcher) nestlings prefer local dialects over the unfamiliar dialects of nonlocal *F. hypoleuca* populations: the more begging calls the nestlings made in response to a song, the stronger their preference. The researchers found that nestlings produced more begging calls in response to their own dialect than to nonlocal dialects. Since song preference plays a role in songbird mate selection, the finding suggests that _____ blank

Which choice most logically completes the text?

- A. *F. hypoleuca* nestlings' preference for their own dialect likely disappears as they mature to promote socialization between different *F. hypoleuca* populations.
- B. *F. hypoleuca* nestlings who show an early preference for their own dialect are likely to receive more food from their caretakers than nestlings who show no preferences among any *F. hypoleuca* dialects.
- C. *F. hypoleuca* nestlings' preference for their own dialect likely drives them when they mature to reproduce with other *F. hypoleuca* from local rather than nonlocal populations.
- D. *F. hypoleuca* nestlings show a preference for both local *F. hypoleuca* dialects and the songs of other local songbirds over the songs of nonlocal birds of any species.

Zines are small-scale, self-printed magazines. They have been around since the Black literary zine *Fire!!* was created in the 1920s. Since then, zines have appealed to creators looking for an inexpensive form of expression to share with a select audience. Zine creators often mix art with social commentary and challenge mainstream culture. At first, the internet appeared to replace the zine, but this old form persists. Today, there are enough zines in the United States to support annual zine festivals. This suggests that _____ blank

Which choice most logically completes the text?

- A. creators can reach a larger audience by posting online.
- B. zines are still a meaningful form of expression.
- C. creators can continue to explore new art forms.
- D. zines are good sources of mainstream culture.

The Cretaceous pterosaur *Tupandactylus navigans* is known for having an anomalously oversized head crest. Until an almost complete fossil skeleton was found in Brazil, paleontologists had been able to study only skull specimens from *T. navigans*, though it was presumed that, like other pterosaurs, the species's primary form of locomotion was powered flight. Examining the fuller skeleton in 2016, Victor Beccari and his team determined that *T. navigans* had long hind legs, short wings, and an unusually long neck—characteristics that, combined with the creature's large-crested head, would have made sustained flight difficult and walking upright relatively comfortable. Based on these findings the team suggests that *T. navigans* likely _____ blank

Which choice most logically completes the text?

- A. flew for longer distances than did other pterosaur species that had oversized head crests.
- B. had longer wings than other pterosaur species considered to have been comfortable walking.
- C. had a smaller head than researchers expected based on the earlier *T. navigans* skull specimens.
- D. flew for shorter distances and spent more time walking than researchers previously thought.

Many mosquito repellents contain natural components that work by activating multiple odor receptors on mosquitoes' antennae. As the insects develop resistance, new repellents are needed. Ke Dong and her team found that EBF, a molecular component of a chrysanthemum-flower extract, can repel mosquitoes by activating just one odor receptor—and this receptor, Or31, is present in all mosquito species known to carry diseases. Therefore, the researchers suggest that in developing new repellents, it would be most useful to _____ blank

Which choice most logically completes the text?

- A. identify molecular components similar to EBF that target the activation of Or31 receptors.
- B. investigate alternative methods for extracting EBF molecules from chrysanthemums.
- C. verify the precise locations of Or31 and other odor receptors on mosquitoes' antennae.
- D. determine the maximum number of different odor receptors that can be activated by a single molecule.